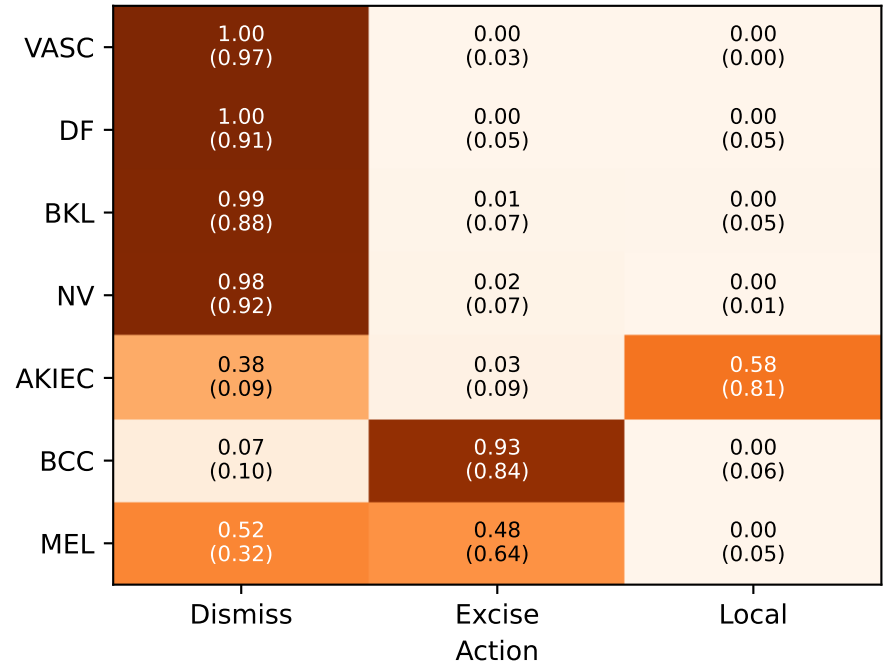
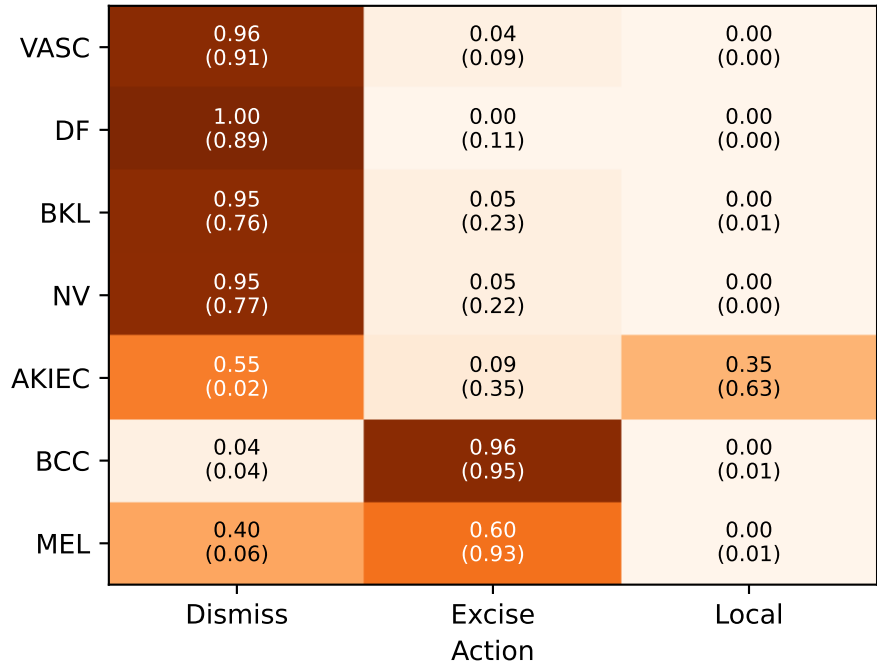


Lesion-level Management Replication: 3-action DQN vs SL baselines vs Paper Fig 2

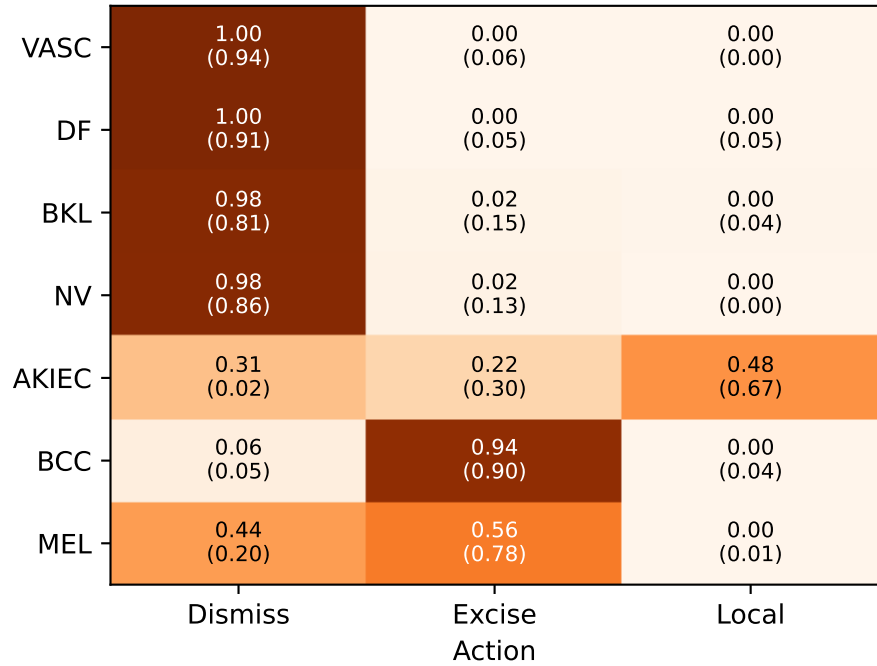
Naive SL (ours / Fig 2f)



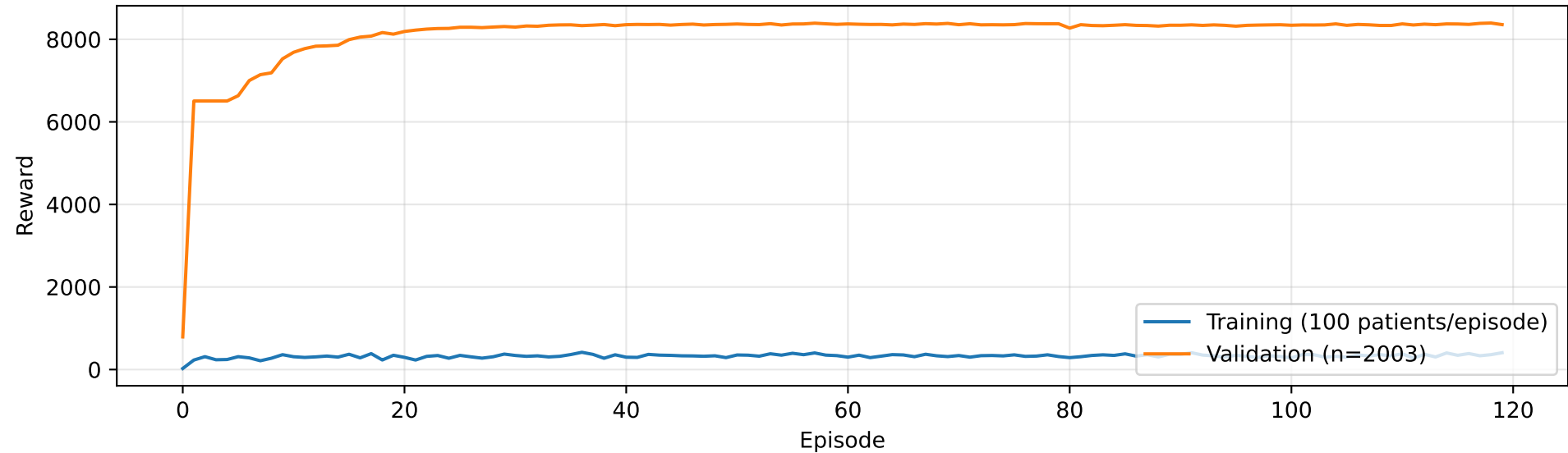
Threshold-SL (ours / Fig 2g)



RL DQN (ours / Fig 2h)



RL DQN reward over training



Key safety metrics

Approach	MEL→Excise	NV→Excise	AKIEC→Local
Naive (ours)	0.48	0.02	0.58
Naive (paper)	0.64	0.07	0.81
Thr-SL (ours)	0.60	0.05	0.35
Thr-SL (paper)	0.93	0.22	0.63
RL DQN (ours)	0.56	0.02	0.48
RL DQN (paper)	0.78	0.13	0.67

Take-aways: (1) RL DQN beats Naive on melanoma safety (+8 pp) and adds AKIEC→cryo routing the Naive cannot produce. (2) Threshold-SL slightly outperforms RL on MEL→excise in our run (0.60 vs 0.56), unlike paper where they tie ~0.78-0.93. (3) Magnitudes are below paper because (a) we used internal HAM10000 80/20 split, not the external ISIC 2018 set, and (b) our upstream SL classifier is less confident on MEL than the paper's. (4) RL convergence is fast — validation reward plateaus by ~episode 30, justifying shorter training in future runs.